Introduction

From this chapter, we are turning ourselves towards learning about second kind of statistics i.e. "Statistical Inference". Statistical inference is an art of making decisions in uncertain situations. The whole art of statistical inference is based on two concepts. One is "Probability Theory" and the other is "Sampling Theory". We are going to discuss both the concepts before the discussion on the statistical inference.

The word probability or chance is not new for every one of us. Hardly a day goes by in which we do not make statements which involve probability. For example, "What are the chances that it will rain today?", "I will probably go abroad this year.", "What are the chances of Pakistan's winning in the game?", "What are the chances of getting head on upper side of the coin?". All these statements and thousands like these indicate some sort of uncertainty. The probability theory helps us in measuring numerically such sort of uncertainty. So probability can be defined as:

"Probability is the measurement of uncertainty."

OR

"Probability is a measure of the likelihood (chance) of something happening."

The foundations of probability were laid in connection with gambling problems. Today the probability has a wider field of application and is used to make decisions in Economics, Management, Sociology, Astronomy, Physics and Engineering where risk and uncertainty is involved.

BASIC CONCEPTS RELATING WITH PROBABILITY THEORY

1. Experiment

In statistics, the term experiment is used in much wider sense than in physics or chemistry. It can be defined as:

A planned activity or process which generates a set of data is called Experiment.

A single performance of an experiment is called a Trail. The result obtained from an experiment or trail is called an outcome.

Examples

1. Playing a game
2. Measuring temperature at different places of a city
3. Counting the number of students in classes
4. A consumer is asked which of the two products he prefers.

2. Random Experiment

An experiment which produces different results/outcomes when it is repeated under similar conditions is called Random Experiment.

A random experiment always has two or more possible outcomes. An experiment that has only one possible outcome is not a random experiment. Also, the outcomes of a random experiment cannot be predicted in advance.

Examples

1. Tossing a coin
2. Throwing a die
3. Selecting a bolt from a lot and observing whether it is defective or not

3. Sample Space

A set consisting of all possible outcomes of a random experiment is called Sample Space.

It is usually denoted by S. The elements of the sample space i.e. the possible outcomes are called the Sample Points.

Examples

> **Remember:**
> Tossing two coins once and tossing a coin two times are equivalent. Both will have the same sample space.

1. R.E: Tossing a coin
   The sample space is: $S = \{H, T\}$
2. R.E: Tossing two coins
   The sample space is: $S = \{HH, HT, TH, TT\}$
3. R.E: Tossing three coins
   The sample space is: $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$
4. R.E: Rolling a die
   The sample space is: $S = \{1, 2, 3, 4, 5, 6\}$
5. R.E: Throwing two dice
   The sample space is:

$$S = \left\{\begin{matrix} (1,1), (1,2), (1,3), (1,4), (1,5), (1,6) \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6) \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6) \\ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6) \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6) \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6) \end{matrix}\right\}$$

4. Event

Any subset of the sample space is called Event.

Event is a group of one or more outcomes. Events are usually denoted by first capital letters of English alphabet like A, B, C etc.

Examples

1. R.E: Tossing a coin
   The sample space is: $S = \{H, T\}$
   Some possible events of S are:
   $A = \{H\}$, $B = \{T\}$ and $C = \{H, T\}$.
2. R.E: Tossing two coins
   The sample space is: $S = \{HH, HT, TH, TT\}$

Some possible events of S are:
A = $\{HH, HT, TH\}$ and B = $\{TT\}$.

5. Mutually Exclusive Events

Two events are said to be mutually exclusive or disjoint events if they cannot occur together in a single trail.

Mathematically, two events A and B are said to be mutually exclusive events if they have no elements in common.

i.e. $A \cap B = \phi$

If events are mutually exclusive, the occurrence of one event prevents the occurrence of the other event.

Examples

1. Head and tail when a coin is tossed
2. Even and odd number when a die is rolled
3. Life and death
4. Fail and pass

6. Not-Mutually Exclusive Events

Two events are said to be not-mutually exclusive events if they can occur together in a single trail.

Mathematically, two events A and B are said to be not-mutually exclusive if they have some elements in common.

i.e. $A \cap B \neq \phi$

1. Watching TV and eating something
2. Walking and talking to someone
3. Rain and sunshine
4. Winning election for different constituencies in an election
5. Passing different papers in a single exam
6. A card drawn from a pack of 52 cards may be a red card and a diamond card

7. Exhaustive Events

Two or more than two events defined in same sample space are said to be collectively exhaustive if their union generates the complete sample space.

i.e. if $A \cup B = S$, then A and B are collectively exhaustive.

For example, in tossing two coins, the events A = $\{HH, HT, TH\}$ and B = $\{TT\}$ are collectively exhaustive because $A \cup B$ = $\{HH, HT, TH\} \cup \{TT\}$ = $\{HH, HT, TH, TT\}$ = S.

8. Equally Likely Events

Events which have the same chance of occurrence are called Equally Likely Events.

For example, when a fair coin is tossed, the chances of occurrence of head and tail are equal so they are equally likely events. Similarly, the occurrence of even and odd number when a fair die is rolled is equally likely events.

9. Complementary Events

The complementary event of an event A is the event which consists of all the sample points in the sample space which are not in the event A.

Complementary events are opposite to each other. Complementary event of an event A is denoted by A^c or $\overline{A}$.

For example, in the experiment of rolling a die, if A = $\{1, 3, 5\}$ then its complementary event will be: $\overline{A} = \{2, 4, 6\}$.

The sum of the probabilities of an event and its complementary event is always equal to 1.

i.e. $$P(A) + P(A^c) = 1$$

$$\Rightarrow \quad P(A) = 1 - P(A^c) \quad \text{or} \quad P(A^c) = 1 - P(A)$$

Note:
Complementary events are always mutually exclusive as well exhaustive.

Note:
$P(A) = 1 - P(A^c)$
or $P(A^c) = 1 - P(A)$
It is called Law of Complementation.

10. Sure Event

An event which is sure to occur is called Sure Event.

It is denoted by S.
For example, if a die is thrown then the event ‘obtaining a number less than 7’ is a sure event.

11. Impossible Event

An event which cannot occur is called Impossible Event.
It is denoted by ϕ. For example, if a die is thrown then the event ‘obtaining a number 7’ is an impossible event.

Rules of Counting

When the number of sample points in a sample space or in an event becomes large, it becomes very difficult to list them all and count the number of sample points in the sample apace or in an event. We than need some methods or rules which help us to count the number of sample points without actually listing them. These rules are described below.

1. Rule of Multiplication/Fundamental Counting Rule

It says that when there are **m** ways of doing something and **n** ways of doing another and these are independent of each other then there are **m × n** ways of doing both.

Multiplication rule can be extended to more than two operations. For example, if there are three events **m, n** and **p**, then number of ways that all events can occur is **m × n × p.**

Two events being independent of each other means that one event does not impact the other.

2. Rule of Permutation

Permutation can be defined as:

An arrangement of objects in a specific order is called Permutation.

It is a group of items where order does matter. The number of permutations of 'n' distinct objects selected 'r' at a time is denoted by ${}^{n}P_{r}$ and is given by:

$$ {}^{n}P_{r} = \frac{n!}{(n-r)!} $$

Example:
How many words can be formed from the letters of the word "SAMPLE" if:

a) Four letters are taken.
b) All the letters are taken.

Example:
In how many ways can five different ornaments be arranged in a shelf?

Example:
A club consists of four members. How many sample points are in the sample space when three officers: president, secretary and treasurer, are to be chosen.

Example:
How many four digit numbers can be formed from the digits 1, 2, 3, 4, 5 and 6 if each digit cannot repeat more than once.

Permutations when Some Objects are Similar

The number of permutations of n objects of which n_1 are of one kind, n_2 are of second kind, … and n_k are of k^{th} kind is given by:

$$ P = \frac{n!}{n_1!\ n_2!\ n_3!\ \dots\ n_k!} $$

Example:
How many words can be formed from the letters of the word "STATISTICS".

Example:
How many different permutations are possible of 7 identical red books and 5 identical green books?

Example:
How many different ways can 6 objects be arranged if 3 of them are identical?

3. Rule of Combination

Combination can be defined as:

An arrangement of objects ignoring the order is called Combination.

It is a group of items where order does not matter. The number of combinations of 'n' distinct objects selected 'r' at a time is denoted by ${}^{n}C_{r}$ or $\binom{n}{r}$ and is given by:

$$ {}^{n}C_{r} = \frac{n!}{r!\,(n-r)!} $$

Note:
Permutation = Ordered Combination

The above formula is also called **Binomial Coefficient**.

Example:
In how many ways a committee of three persons can be formed from four persons.

Example:
In how many ways 5 cards can be selected from a deck of 52 cards.

Definition of Probability

Let 'n' be the total number of possible mutually exclusive and equally likely outcomes out of which 'm' be the number of favorable outcomes for an event 'A', then the probability of the event A, denoted by P(A), is defined as:

$$ P(A) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} $$

$$ = \frac{m}{n} $$

Note:
This is called Classical definition of probability.

Some Important Points about Probability

1. The probability of an event always lies between 0 and 1 (both inclusive) or in percentage-between 0% and 100%.
   i.e. $0 \le P(A) \le 1$
2. The probability of an impossible event is 0.
   i.e. $P(\phi) = 0$
3. The probability of a sure event is 1.
   i.e. $P(S) = 1$

Remember:
The probability of an event can never be negative and greater than 1.

Addition Law of Probability for Mutually Exclusive Events

This law states that:

"If A and B are two mutually exclusive events, then the probability that event A or B occurs is equal to the sum of the separate probabilities of A and B occurring."

Symbolically, it can be written as:

$$ P(A \cup B) = P(A) + P(B) $$

Addition Law of Probability for Not-Mutually Exclusive Events

This law states that:

"If A and B are any two events, then the probability that event A or B or both occur is equal to the sum of the separate probabilities of A and B occurring minus the probability that they both occur together."

Symbolically, it can be written as:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Conditional Probability

The probability found on the basis of some extra information is called Conditional Probability.

OR

The probability of the occurrence of an event when it is known that some other event has already occurred is called Conditional Probability.

The effect of such extra information is to reduce the sample space by excluding some outcomes as being impossible. So conditional probability can also be defined as "the probability calculated on the basis of a reduced sample space is called Conditional Probability."

It is denoted by $P(A/B)$ and is read as "the probability that A occurs given that B has occurred" or simply "the probability of A given B." It is defined by:

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Similarly,

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

Independent and Dependent Events

Independent Events

Two events are independent if the outcome of one does not affect the outcome of the other.

OR

When the occurrence of one event does not affect the probability of the occurrence of other event, such events are called Independent Events.

For example, if a dice is rolled twice the number on the second roll is independent of the number on the first roll.

Dependent Events

When the occurrence of one event affects the probability of the occurrence of other event, such events are called Dependent Events.

In case of sampling with replacement, the events are independent while in case of sampling without replacement, the events are dependent.

Multiplication Law of Probability for Independent Events

This law states that:

"If A and B are two independent events, then the probability that both of them occur is equal to the product of their separate probabilities."

That is: $$P(\text{A and B}) = P(A) \times P(B)$$

or $$P(A \cap B) = P(A) \times P(B)$$

Multiplication Law of Probability for Dependent Events

This law states that:

"If A and B are any two events, then the probability that both A and B occur is equal to the probability that A occurs multiplied by the conditional probability that B occurs given that A has already occurred."

i.e. $$P(A \cap B) = P(A)\ P(B/A)$$

or $$P(A \cap B) = P(B)\ P(A/B)$$

Note:

The following MCQ's in the ICAP Study Text are related to this chapter.

11.1, 11.3 to 11.7, 11.9 to 11.55, 11.57 to 11.66, 11.68.

Questions of ICAP Study Text Related to Rules of Counting (Pages: 268 to 272)

1) A lady has 10 pairs of shoes and 12 dresses. How many different possible combinations are to wear shoes and dress?
Ans: 120

2) A car manufacturer sells a model of car with the following options:

Body style:	Sedan, hatchback, soft top	3
Colors:	Choice of 15	15
Internal trim:	3 colors of leather and 6 colors of fabric	9
Engine size:	1.8l, 2.2l and 2.5l turbo	3

How many different variations of the car are offered by the car manufacturer?
Ans: 1215

3) In how many ways the three letters (A, B and C) be arranged.
Ans: 6

4) Qadir has 4 sisters who ring him once every Saturday. How many possible arrangements of phone calls could there be?
Ans: 24

5) In how many ways can 8 objects be arranged?
Ans: 40320

6) There are 9 books on a shelf. The librarian will log three of these first in order. How many ways does librarian have?
Ans: 504

7) There are 10 balls in a bag. Each ball is carrying a different number from 1 to 10. Three balls are selected at random from the bag and not replaced before the next draw is made. How many different **permutations** are possible?
Ans: 720

8) 6 friends visit a cinema but can only buy 4 seats together. How many ways could the 4 seats be filled by the six friends?
Ans: 360

9) A military code is constructed where each code is represented by 4 randomly selected numbers in the sequence 1 to 50. How many permutations are possible for a single code if repetition is allowed?
Ans: 6250000

10) How many possible arrangements are there of the three letters ABB?
Ans: 3

11) How many different ways can 7 objects be arranged if 4 of them are identical?
Ans: 210

12) How many ways can the letters in PAKISTAN be arranged?
Ans: 20160

13) How many ways can the letters in COMMITMENT be arranged?
Ans: 302400

14) There are 10 balls in a bag each ball carrying a different number from 1 to 10. Three balls are selected at random from the bag and not replaced before the next draw is made. How many different combinations are possible?
Ans: 120

15) A university hockey team has a squad of 15 players. The coach carries out a strict policy that every player is chosen randomly. How many possible teams of 11 can be picked?
Ans: 1365

16) Five students from an honors' class of 10 are to be chosen to attend a competition. One must be the head boy. How many possible teams are there?
Ans: 126

Questions of ICAP Study Text

1) **Example**
In tossing a coin, what is the probability of rolling a head?
Ans: 1/2

2) **Example**
In rolling a dice, what is the probability of rolling a 5? What is the probability of rolling an even number?
Ans: 1/6, 3/6

3) **Example**
A box contains 10 pens (4 Blue, 2 Red, 3 Black and 1 Green).

What is the probability of picking a blue pen?

What is the probability of not picking a black pen?

Ans: 0.4, 0.7

4) **Example**
If two dice are rolled, what is the probability of rolling two 6s?
Ans: 1/36

5) **Example**
There are 4 white and 5 blue pens. What is the probability that two white pens are drawn respectively if none of the pens are replaced?
Ans: 1/6

6) **Example**
A person is dealt two cards from a pack. What is the probability that both are aces?
Ans: 0.0045

7) **Example**
There are 8 blue pens and 7 black pens in a box. If the pens are drawn six times without replacing the pens, what is the probability that exactly three blue pens are drawn?
(pg # 304)

8) **Example**
What is the probability of there being exactly two red cards in the five selected from a deck?
Ans: 0.325
(pg # 303)

9) **Example**
What is the probability of rolling a 3 or a 4 when rolling a dice?
Ans: 1/3

10) **Example**
What is the probability of selecting a jack or a queen from a normal pack of 52 cards?
Ans: 2/13

11) Example

What is the probability of selecting a club or a spade from a normal pack of 52 cards?
Ans: 1/2

12) Example

What is the probability of selecting an ace or a heart from a normal pack of 52 cards?
Ans: 16/52

13) Example

Two dice are rolled- 1 red and 1 blue. What is the probability of rolling a 6 on the red or blue dice?
Ans: 11/36

14) Example

A box contains 10 pens (4 Blue, 2 Red, 3 Black and 1 Green), what is the probability that a pen is blue or black?

Ans: 0.7
(pg # 275)

15) Example

The chances of survival of two accident victims are 0.3 and 0.6. What is the probability that both of them will survive?
(pg # 276)

16) Example

A box contains 4 Blue, 2 Red, 3 Black and 1 Green pen. What is the probability that both pens are blue (a) If the first pen is replaced prior to selecting the second. (b) If the first pen is not replaced prior to selecting the second.

Ans: (a) 16/100 (b) 12/90

(pg # 278)

17) Example

Three dice are rolled. What is the probability of getting at least one six?

(pg # 276)

18) Example

Daily sales of a firm and probability of achieving the sales is provided below

Daily sales	**P**
0	0.2
1	0.3
2	0.25
3	0.15
4	0.10
Total	**1.0**

What is the probability of daily sales of more than 2?
What is the probability of daily sales being at least 1?
What is the probability of total sales of 6 in two consecutive days?
(pg # 275)

19) Example

An item is made in three stages. At the first stage, it is formed on one of the four machines A, B, C or D with equal probability. At the second stage, it is trimmed on one of the three machines E, F or G with equal probability. Finally, it is polished on one of two polishers H and I, and is twice as likely to be polished on the former as this machine works twice as quickly as the other. What is the probability that an item is:

a) Polished on H?
b) Trimmed on either F or G?
c) Formed on either A or B, trimmed on F and polished on H?
d) Either formed on A and polished on I or formed on B and polished on H?
e) Either formed on A or trimmed on F?

(pg # 276)

20) Example

Three people A, B and C get into a lift on the ground floor of a building. The building has 6 floors comprising the ground floor and floors 1 to 5. The probability that a person entering the lift at the ground floor requires a given floor is as follows:

Floor required	Probability
1	0.1
2	0.2
3	0.2
4	0.3
5	0.2

a) Calculate the probability that A requires either floor 3, 4 or 5.
b) Calculate the probability that neither A, B nor C requires floor 1.
c) Calculate the probability that A and B require the same floor.
d) Calculate the probability that A and B require different floors.
e) Calculate the probability that only one of A, B and C requires floor 2.
f) Calculate the probability that A travels three floors further than B.
(pg # 277)

21) Example

Sindh Export Limited – Employee Report

Department	Male		Female		Total
	Under 18	18 and over	Under 18	18 and over	
Senior staff	0	15	0	5	**20**
Junior staff	6	4	8	12	**30**
Transport	2	34	2	2	**40**
Stores	1	20	2	17	**40**
Maintenance	1	3	4	12	**20**
Production	0	4	24	72	**100**
Total	**10**	**80**	**40**	**120**	**250**

(a) All employees are included in a weekly draw for a prize. What is the probability that in one particular week the winner will be:

(i) a male.
(ii) a member of junior staff.
(iii) a female member of staff.
(iv) from the maintenance or production departments.
(v) a male or transport worker?

(b) Two employees are to be selected at random to represent the company at the annual Packaging Exhibition in London. What is the probability that:

(i) both are female.
(ii) both are production workers.
(iii) one is a male under 18 and the other a female 18 and over?

(pg # 279)

Questions of PAC Study Text

1) **Class Work-2:**
In how many ways a man can wear a suit and a tie if the man has 3 suits and 2 ties.

2) **Class Work-1:**
In how many ways a coin and die can be thrown together.

3) **Class Work-16:**
If two dice are thrown, what is the probability that:

i) Sum 8 appears.
ii) Sum more than 10 appears.
iii) Sum divisible by 4.
iv) Sum is odd.
v) At least one ace appears.
vi) Two shown the same numbers (doublet).
vii) Product of numbers divisible by 4.
viii) An odd number on one die and multiple of 3 on other die.

4) **Class Work-11:**
What is the probability that if a card is drawn at random from an ordinary pack of cards, it is: (i) a red card (ii) a club?

5) **Class Work-21:**
A card is drawn from a pack of 52 cards. What is the probability that it is:
(i) an ace or king.
(ii) an ace or diamond card.
(iii) a king or black card.

Ans: (i) 8/52 (ii) 16/52 (iii) 28/52

6) **Class Work-23:**
Two dice are thrown, what is probability of getting:
(i) Sum 7 or 11.
(ii) Sum 8 or doublet.

Ans: (i) 8/36 (ii) 10/36

7) **Class Work-28:**
A dice is rolled. What is the probability of getting a number greater than 3 given that it is an even number?

8) <u>Class Work-30</u>:

A sample survey of 120 tubes manufactured by two companies A and B gave the following results.

	Company A	**Company B**	**Total**
Defective	5	10	**15**
Non defective	55	50	**105**
Total	**60**	**60**	**120**

(i) If a tube manufactured by company A is selected, what is the probability that it is defective?

(ii) If a defective tube is selected, what is probability that it is manufactured by company A?

9) <u>Class Work-33</u>:

A small town has one fire engine and one ambulance available for emergencies. The probability that fire engine is available when needed is 0.98 and the probability that ambulance is available when needed is 0.92. In the event of an injury from a burning building, find the probability that both the fire engine and ambulance will be available.

Ans: 0.9016

10) <u>Class Work-31</u>:

A box contains 5 black and 3 white balls. If two balls are drawn at random without replacement, what is the probability that 1^{st} is black and 2^{nd} is white.

Ans: 15/56

11) <u>Class Work-34</u>:

The probability that A will be alive after 10 years to come is 5/7 and for B it is 7/9. Find the probability that:

(i) Both of them will die.
(ii) A alive and B dead.
(iii) B will be alive and A dead.
(iv) Both will be alive.
(v) At least one will be alive.

Ans: (i) 4/63 (ii) 10/63 (iii) 14/63 (iv) 35/63 (v) 59/63